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/*
Programmed by: Thomas M. Grover
Programmed on: 2025-04-08
Programmed to: Complete FP Phase 1
Modified by: N/A
Modified on: N/A
Modified to: N/A
*/

x 'cd L:\st445\Data';
libname Fmts '.';

x 'cd L:\st445\Data\BookData\BeverageCompanyCaseStudy';
libname InputDS '.';
filename InputRaw '.';
libname axDB2016 access "2016data.accdb";

x 'cd L:\st445\Results';
libname Results '.';

x 'cd S:\Documents\FP01';
libname Exam '.';

proc format library = Exam cntlin = Exam.SodaNames;
    value SodaNames 1  = 'Cola'
                    2  = 'Diet Cola'
                    3  = 'Cherry Cola'
                    4  = 'Diet Cherry Cola'
                    5  = 'Vanilla Cola'
                    6  = 'Diet Vanilla Cola'
                    7  = 'Professor Zesty'
                    8  = 'Diet Professor Zesty'
                    9  = 'Citrus Splash'
                    10 = 'Diet Citrus Splash'
                    11 = 'Lemon-Lime'
                    12 = 'Diet Lemon-Lime'
                    13 = 'Orange Fizzy'
                    14 = 'Diet Orange Fizzy'
                    15 = 'Grape Fizzy'
                    16 = 'Diet Grape Fizzy';
run;

ods listing close;
options fmtsearch = (Exam Fmts);


*Read Counties and Make a Local Copy;
data Exam.Counties;
  set axDB2016.counties(drop = region
                        rename = (state=StateFIPS county=CountyFIPS));
run;
libname axDB2016 clear;

*Read NonColaSouth;
data Exam.NonColaSouth;
    attrib ProductName length = $50;
    infile InputRaw('Non-Cola--NC,SC,GA.dat') firstobs = 7;
    input StateFips     1-2
          CountyFips    3-5
          ProductName $ 6-25
          DummySize        $ 26-35
          UnitSize      36-38
          @39 Date      mmddyy10.
          UnitsSold     49-55;
run;
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*Read NonColaNorth;
data Exam.NonColaNorth;
    infile InputRaw('Non-Cola--DC-MD-VA.dat') firstobs = 5 dlm = '   ';
    input StateFips 2.
          CountyFips 3.
          ProductCode $15.
          DummyDate : $20.
          UnitsSold : 8.;
run;

*Read EnergySouth;
data Exam.EnergySouth;
    attrib ProductName length = $50.;
    infile InputRaw('Energy--NC,SC,GA.txt') firstobs = 2
                                            dlm = '09'x
                                            truncover;
    input StateFips
          CountyFips
          ProductName : $20.
          DummySize $
          UnitSize
          Date date10.
          UnitsSold;
run;

*Read EnergyNorth;
data Exam.EnergyNorth;
    infile InputRaw('Energy--DC-MD-VA.txt') firstobs = 2
                                            dlm = '09'x;
    input StateFips
          CountyFips
          ProductCode : $15.
          DummyDate : $20.
          UnitsSold;
run;

*Read OtherSouth;
data Exam.OtherSouth;
    attrib ProductName length = $50.;
    infile InputRaw('Other--NC,SC,GA.csv') firstobs = 2
                                           dsd
                                           truncover;
    input StateFips
          CountyFips
          ProductName ~ $20.
          DummySize $
          UnitSize
          Date date10.
          UnitsSold;
run;

*Read OtherNorth;
data Exam.OtherNorth;
    infile InputRaw('Other--DC-MD-VA.csv') firstobs = 2
                                           dsd;
    input StateFips
          CountyFips
          ProductCode : $15.
          DummyDate ~ $20.
          UnitsSold;
run;
```

```
*Concatenating Cola, Non-Cola, Energy, and Other Datasets from Both Regionss;
data Exam.AllDrinks(drop = ProductCode code NameCode DummySize DummyDate);
    attrib StateFips          label = 'State FIPS'
                              format = best12.
           CountyFips         label = 'County FIPS'
                              format = best12.
           Region             label = 'Region'
           ProductName        label = 'Beverage Name'
                              length = $50.
           Type               label = 'Beverage Type'
                              length = $8.
           Flavor             label = 'Beverage Flavor'
                              length = $50.
           ProductCategory    label = 'Beverage Category'
                              length = $14.
           ProductSubCategory label = 'Beverage Sub-Category'
           Size               label = 'Beverage Volume'
           UnitSize           label = 'Beverage Quantity'
                              format = best12.
           Container          label = 'Beverage Container'
           Date               label = 'Sale Date'
                              format = date9.
           UnitsSold          label = 'Units Sold'
                              format = comma7.;
    set InputDS.ColaNCSCGA(in=CS)
        InputDS.ColaDCMDVA(in=CN)
        Exam.NonColaSouth(in=NCS)
        Exam.NonColaNorth(in=NCN)
        Exam.EnergySouth(in=ES)
        Exam.EnergyNorth(in=EN)
        Exam.OtherSouth(in=OS)
        Exam.OtherNorth(in=ON);
    if CN+NCN+EN+ON=1 then Date = input(strip(DummyDate),??date9.);
        else Date=Date;
    NameCode = 1*CN + 2*NCN + 3*EN + 4*ON;
    select(NameCode);
         when(1)   ProductName = put(input(substr(code,3,1),best.),SodaNames.);
         when(2)   ProductName = put(input(compress(substr(ProductCode,3,2),,'dk'),best.),SodaNames.);
         when(3)   ProductName = put(compress(substr(ProductCode,3,2),,'dk'),EnergyNames.);
         when(4)   ProductName = put(substr(ProductCode,3,1),OtherNames.);
         otherwise ProductName = propcase(ProductName);
    end;
    if index(DummySize,'ounces') ge 1 then Size = tranwrd(Dummysize,'ounces','oz');
        else if index(DummySize,'ounce') ge 1 then Size = tranwrd(DummySize,'ounce','oz');
            else if index(DummySize,'liters') ge 1 then Size = tranwrd(DummySize,'liters','liter');
    if CS+NCS+ES+OS=1 then Size = lowcase(size);
        else if CN = 1 then Size = scan(code,3,'-');
            else Size = scan(ProductCode,3,'-');
    if CS+NCS+ES+OS = 1 then Region = 'South';
        else if CN+NCN+EN+ON = 1 then Region = 'North';
    if index(ProductName,'Diet') ge 1 then Type = 'Diet';
        else Type = 'Non-Diet';
    if CS+CN=1 then ProductCategory = 'Soda: Cola';
        else if NCS+NCN=1 then ProductCategory = 'Soda: Non-Cola';
            else if ES+EN=1 then ProductCategory = 'Energy';
                else if index(ProductName,'Nutrition') ge 1 then ProductCategory = 'Nutritonal Water';
                    else ProductCategory = 'Non-Soda Ades';
    if index(ProductName,'Mega Zip') ge 1 then ProductSubCategory = 'Mega Zip';
        else if index(ProductName,'Big Zip') ge 1 then ProductSubCategory = 'Big Zip';
            else if index(ProductName,'Zip') ge 1 then ProductSubCategory = 'Zip';
    if CS+CN+NCN+NCS=1 then Flavor = strip(tranwrd(ProductName,'Diet',''));
        else if index(ProductName,'Grape') ge 1 then Flavor = 'Grape';
            else if index(ProductName,'Berry') ge 1 then Flavor = 'Berry';
                else if index(ProductName,'Orange') ge 1 then Flavor = 'Orange';
                    else if index(ProductName,'Lemonade') ge 1 then Flavor = 'Lemonade';
                        else Flavor = 'Orangeade';
    if CS+NCS+ES+OS=1 then UnitSize = UnitSize;
        else if CN=1 then UnitSize = scan(code,4,'-');
            else UnitSize = scan(ProductCode,4,'-');
    if index(Size,'liter') ge 1 then Container = 'Bottle';
        else if index(Size,'20') ge 1 then Container = 'Bottle';
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            else Container = 'Can';
run;

proc sort Data = Exam.AllDrinks
          out = Exam.AllSorted;
    by StateFips CountyFips;
run;


*Merging All Drinks With Counties;
data Exam.AllData(drop = _popmeanover1k);
    attrib StateName          label = 'State Name'
                              length = $50.
           StateFips          label = 'State FIPS'
                              format = best12.
           CountyFips         label = 'County FIPS'
                              format = best12.
           CountyName         label = 'County Name'
                              length = $50.
           Region             label = 'Region'
           popestimate2016    label = 'Estimated Population in 2016'
                              format = comma10.
           popestimate2017    label = 'Estimated Population in 2017'
                              format = comma10.
           ProductName        label = 'Beverage Name'
                              length = $50.
           Type               label = 'Beverage Type'
                              length = $8.
           Flavor             label = 'Beverage Flavor'
                              length = $50.
           ProductCategory    label = 'Beverage Category'
                              length = $14.
           ProductSubCategory label = 'Beverage Sub-Category'
           Size               label = 'Beverage Volume'
           UnitSize           label = 'Beverage Quantity'
                              format = best12.
           Container          label = 'Beverage Container'
           Date               label = 'Sale Date'
                              format = date9.
           UnitsSold          label = 'Units Sold'
                              format = comma7.
           SalesPerThousand   label = 'Sales per 1000'
                              format = 7.4;
    merge Exam.AllSorted
          Exam.Counties;
    by StateFips CountyFips;
    _popmeanover1k = (popestimate2016+popestimate2017)/2000;
    if _popmeanover1k > 0 then SalesPerThousand = (UnitsSold/_popmeanover1k);
        else SalesPerThousand = .;
run;

*Output 3.6;
ods output summary = Exam.Activity36;
proc means data = Exam.AllData(where = ((index(ProductName,'Nut') ge 1) and
                                        (Region = 'South') and
                                        (UnitSize = 1)))
           mean median nonobs maxdec=2;
    class ProductName;
    var UnitsSold;
run;

proc sort data = Exam.AllData
          out = Exam.AllDataSorted;
    by Region Type descending UnitsSold;
run;
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*Output 4.4;
ods output summary = Exam.Activity44;
proc means data = Exam.AllDataSorted(where = ((ProductCategory = 'Soda: Cola') and
                                              (Size = '20 oz') and
                                              (UnitSize = 1)))
           nonobs qrange;
    by Region Type;
    class Date;
    var UnitsSold;
run;

*Output Optional;
proc sort data = Exam.AllDrinks(keep = ProductName Type ProductCategory ProductSubCategory Flavor Size Container)
          nodupkey
          out = Exam.OptionalData;
    by ProductCategory ProductSubCategory descending Type Flavor Container;
run;

*Output 5.5;
ods output summary = Exam.Activity55;
proc means data = Exam.AllDataSorted(Where = ((StateName in ('North Carolina', 'South Carolina')) and
                                              (month(date) = 8) and
                                              (Size = '12 oz') and
                                              (UnitSize = 1) and
                                              (Flavor = 'Cola')))
           nonobs sum;
    by Type;
    class Date StateName;
    var UnitsSold;
run;

ods listing;
quit;
```